Ex - 07
Data Visualization Using Power BI

AIM

To visualize data using Power BI by connecting data sources, creating charts, applying DAX calculations, and building an interactive dashboard using the Grains dataset.

PROBLEM STATEMENT

To analyze the Grains dataset using Power BI by creating bar charts, line charts, pie charts, calculated columns, and measures to present meaningful insights such as total sales, grain-wise sales, region-wise sales, and category-wise distribution.

PROCEDURE

1. Open Microsoft Power BI Desktop and explore the Report View, Data View, Model View, and Visualizations pane.
2. Connect to the Grains dataset (Grains_Dataset.csv) using Home > Get Data, select the required file, and click Load to import the data.
3. Connect to a SQL database by selecting Home > Get Data > SQL Server, entering the server details, and selecting the required tables.
4. Open Power Query Editor using Transform Data to remove duplicates, handle missing values, rename columns, and change data types.
5. Create a bar chart by selecting the Clustered Bar Chart visual and dragging Grain to the Y-axis and Sales to the X-axis to show sales of each grain.
6. Create a line chart by selecting the Line Chart visual, placing the Order_Date (Month) field on the X-axis and Sales on the Y-axis to analyze the monthly sales trend.
7. Create a pie chart by selecting the Pie Chart visual, placing Category in the Legend and Sales in Values to display the category-wise sales distribution.
8. Create a calculated column using Table tools > New column and enter a DAX formula to calculate additional values for each row.
9. Create measures using Table tools > New measure and apply DAX functions such as SUM(), DISTINCTCOUNT() and AVERAGE() to calculate totals and counts.
10. Build an interactive dashboard-style report by arranging charts, KPI cards, and slicers, formatting titles and colors, testing filters, and saving the report as a .pbix file.

IMPLEMENTATION

1. Calculated Column (DAX)

```
Sales Category = IF(Grains[Sales] >= 20000, "High Sales", "Low Sales")
```

2. Measure - Total Sales

```
Total Sales = SUM(Grains[Sales])
```

3. Measure - Total Quantity

```
Total Quantity = SUM(Grains[Quantity_kg])
```

4. Measure - Total Orders

```
Total Orders = DISTINCTCOUNT(Grains[Order_ID])
```

Dashboard components:

• Cards - Total Sales, Total Quantity, Total Orders

• Bar chart - Grain sales (Sum of Sales by Grain)

• Column chart - Sales by region

• Line chart - Sales by month

• Pie chart - Sales distribution by category (Cereal, Millet, Pulse)

• Slicer - Filter by region or category

Note: Replace the table and column names with those in your dataset.

OUTPUT

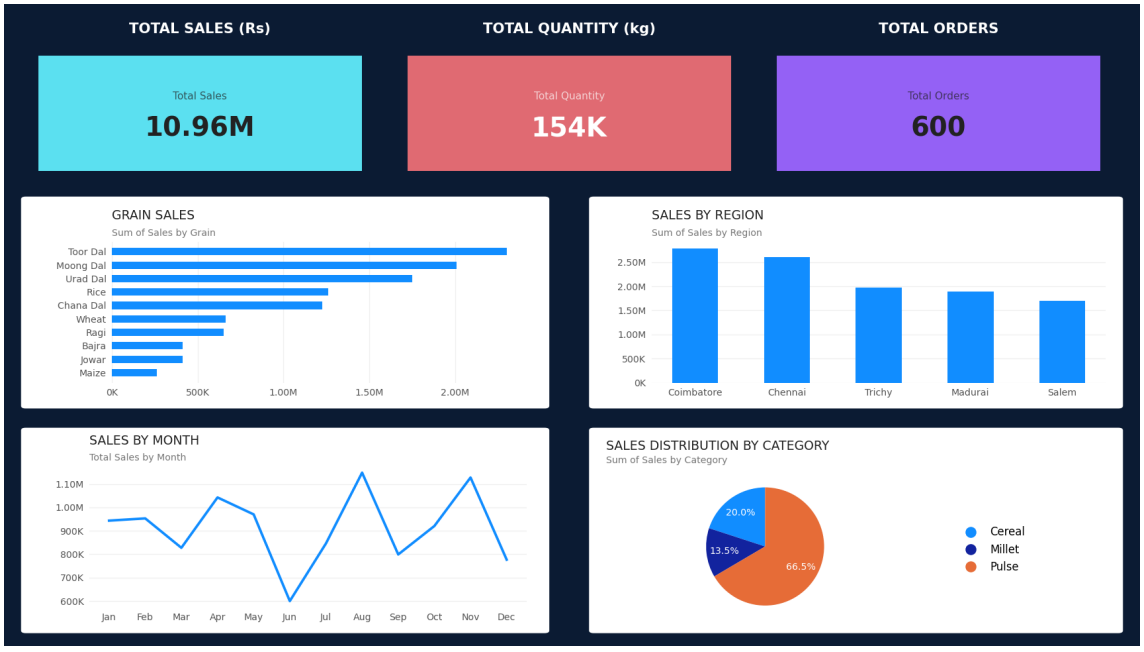


Fig: Grains Sales Analytics Dashboard (Power BI)

RESULT

Data was successfully visualized using Power BI on the Grains dataset. Bar charts, line charts, pie charts, DAX calculated columns, and measures were created and combined into an interactive dashboard for effective data analysis.